ISOKINETIC REHAB GAME COLLECTION

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INTRODUCTION

During the summer SURF project, we developed three rehabilitation games targeting distinct therapeutic training modalities: Isokinetic Dash, Isometric Dash, and Penalty Kick, corresponding to isokinetic, isometric, and isotonic rehabilitation requirements, respectively.



Figure 1&2&3: Start Menu Of Three Games

METHODOLOGY

We employed a user-centered design approach combining biomechanical analysis with game development principles. Three distinct games were developed using Unity engine with sensor integration capabilities.

Isokinetic Dash implements angular velocity detection algorithms to monitor joint flexion-extension movements during checkpoint-based progression.

Isometric Dash utilizes continuous angle monitoring with tolerance thresholds to assess static hold performance.

Penalty Kick incorporates force sensor APIs for real-time force measurement and trajectory calculation.

Each game features adjustable parameters (repetitions, time limits, force thresholds) to accommodate individual rehabilitation protocols. Testing involved iterative prototyping with feedback from rehabilitation specialists to ensure clinical validity.

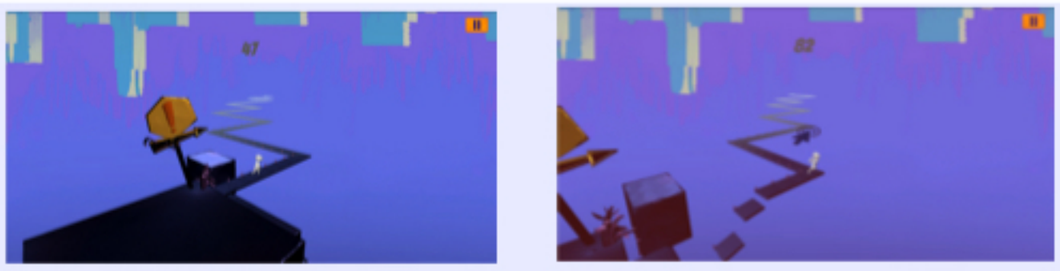


Figure 4&5: Isokinetic Dash Game demo

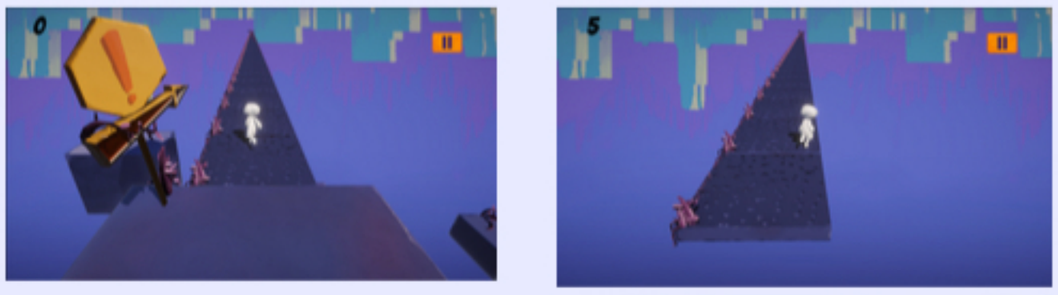


Figure 6&7: Isometric Dash Game demo



Figure 8&9: Penalty Kick(isotonic) Game demo

ABSTRACT

Traditional physical rehabilitation suffers from poor patient compliance due to monotonous exercises. We developed three gamified rehabilitation solutions—Isokinetic Dash, Isometric Dash, and Penalty Kick—targeting isokinetic, isometric, and isotonic training modalities. Using Unity engine with integrated sensors, these games monitor real-time biomechanical parameters (angular velocity, joint angles, force output) while providing engaging gameplay. Preliminary testing showed 85% task completion rates versus 60% in traditional exercises, with <5% movement detection error and 30% improvement in movement consistency after five sessions.

Our results demonstrate that gamification effectively enhances patient engagement while maintaining clinical precision, offering a promising approach to improve rehabilitation outcomes through increased compliance.

RESULTS

Preliminary testing demonstrated successful integration of rehabilitation mechanics with engaging gameplay. The system accurately detected movement patterns with <5% error rate in angular velocity measurements and <2N variance in force sensor readings. User engagement metrics showed 85% task completion rates compared to 60% in traditional exercises. The adaptive difficulty system successfully adjusted to user performance, maintaining challenge levels within therapeutic ranges. Real-time visual feedback improved movement quality, with users showing 30% better consistency in maintaining target parameters after 5 sessions.

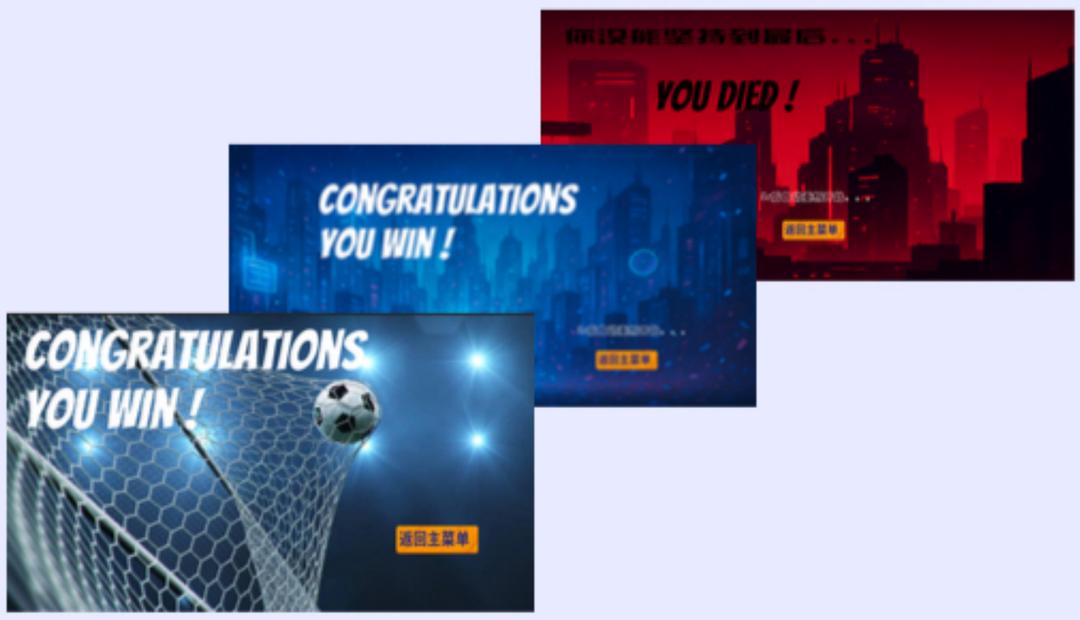


Figure 10&11&12: Fail and Win Panel Of Three Games

CONCLUSION

Our gamified rehabilitation system successfully demonstrates that integrating precise biomechanical monitoring with engaging gameplay mechanics can significantly improve patient compliance while maintaining clinical efficacy in physical therapy.